

Automation & Systems Developer Assignment Tasks

Purpose

This assignment evaluates your ability to understand business problems, design practical systems, identify risks, prioritise work and communicate technical decisions clearly.

No coding or software development is required.

Academic Integrity

Honesty is a core value at Loyola Tutoring. All responses must be your own original work. Responses written or substantially generated using AI without disclosing will be disqualified.

Submission Format

Single compiled PDF only. File name must be Firstname_Lastname(Automation).pdf

Submission Requirements

Item	Requirement
Recommended effort	2–3 hours
Maximum length	6 pages or 8 presentation slides
Required visual	One simple workflow or architecture diagram
Approach	Clearly state assumptions and explain the reasoning behind recommendations
AI use	Permitted, but must be disclosed; candidates must be able to explain and defend every part of the submission

1 Loyola LMS: MVP and Delivery Plan

Loyola Tutoring wants an LMS for administrators, tutors, parents and students. The future system may include:

- Enrolment and class allocation
- Learning materials and recordings
- Homework delivery and submission
- Attendance and progress tracking
- Parent reports
- Auto-marking
- AI-assisted feedback
- Dashboards and reporting

Prepare a practical plan covering:

1. What you would include in the first LMS beta.
2. What you would postpone and why.
3. Whether you would build a custom system, customise an existing LMS or use a hybrid approach.
4. The platforms, software and services you would recommend.
5. A simple high-level architecture.
6. A realistic timeline for the main stages.
7. The people, information and resources you would need.
8. The three biggest project risks.
9. What you believe could realistically be delivered during the first 90 days.

Important: Do not simply list technologies. Explain why your approach is appropriate for Loyola's size, resources and business needs.

2 Weekly Parent-Resource Automation

Every week, Loyola needs to send parents the correct class recording, answer key, worksheet and supporting materials. The recipient list changes because of:

- New enrolments
- Cancellations and refunds
- Class transfers
- Late enrolments
- Missing or incorrect contact details
- Parents with multiple children
- Students attending multiple classes

Design a reliable automated process. Your response should explain:

1. Which system should be the source of truth.
2. How enrolment, cancellation, refund and class-transfer changes would be captured.
3. How resources would be matched to the correct class.
4. How the system would decide who should receive each message.
5. How duplicate sends and missed parents would be prevented.
6. How missing information and failed deliveries would be handled.
7. What staff should review before sending.
8. What logs, alerts and audit records should be maintained.
9. How the process should be tested before launch.
10. How Loyola would measure whether the automation is successful.

Required: Include a simple workflow diagram.

3 Automation Priorities and Failure Response

Part A: Quick-win automations

Identify five automation opportunities that could benefit Loyola in areas such as enrolment, timetabling, class delivery, parent communication, tutor coordination, payroll or timesheets, content management and internal communication.

For each opportunity, briefly provide:

- The current problem
- The proposed automation
- Expected benefit
- Estimated effort
- Main risk or dependency

Rank the five opportunities and explain which two you would implement first.

Part B: Failure scenario

Some parents received the wrong answer key, some active families received nothing, cancelled students still received materials, and several parents received duplicate emails.

Explain:

1. What you would do during the first hour.
2. Whether you would stop the automation.
3. How you would identify affected families.
4. How you would safely resend the correct materials.
5. How you would communicate with the Founder, Head of AI, staff and parents.
6. How you would investigate the root cause.
7. What short-term controls you would introduce.
8. What permanent improvements you would recommend.

Conclude with three design principles that could have prevented the incident.

4 AI Question-Generation System

You are tasked with building a learning management system where students can log in and practise questions. You will initially receive 10 approved questions for each section of the examination. The system must use these questions as a starting point and generate additional questions using AI. Explain how you would design and build the system. Your response should cover:

- The programming languages, models, tools and infrastructure you would use
- How the 10 approved questions would be analysed and used
- The internal process for generating new questions
- How answers and worked solutions would be verified
- How you would control difficulty, topic coverage and repetition
- How you would detect ambiguous, incorrect or low quality questions
- How the system would prevent students from receiving unsuitable questions
- How you would monitor quality after launch

What Loyola is Looking For

- Ask sensible questions rather than make unsupported assumptions.
- Distinguish the MVP from future features.
- Consider failures, exceptions and incorrect data.
- Recommend practical rather than unnecessarily complex solutions.
- Explain trade-offs clearly.
- Prioritise measurable business value.
- Use AI only where it is genuinely appropriate.
- Provide realistic timelines.